Week 3 CUSF Controls Introductory Project - Linearisation and PID

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Part 1: Linearisation methods

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Motivating Example - Inverted Pendulum on a Cart

Horizontally (cart):

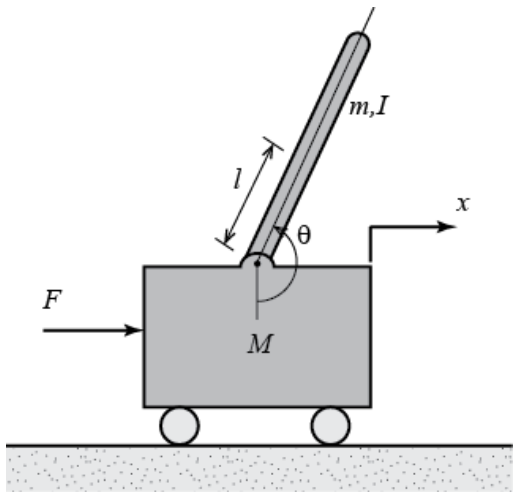
$$M\ddot{x} + N = F \tag{1}$$

Horizontally (rod):

$$N = m\ddot{x} + ml\ddot{\theta}\sin\theta - ml\dot{\theta}^2\sin(\theta) \tag{2}$$

(2) $\rightarrow$ (1):

$$(M+m)\ddot{x} + ml\ddot{\theta}\cos(\theta) - ml\dot{\theta}^2\sin(\theta) = F(t) \tag{3}$$



Continued

Considering the torque on the rod:

$$Pl\sin(\theta) - Nl\cos(theta) - I\ddot{\theta} = 0 \tag{4}$$

Forces perpendicular to rod:

$$N\cos(\theta) - P\sin(\theta) - ml\ddot{\theta} - m\ddot{x}\cos(\theta) + mg\sin(\theta) = 0 \tag{5}$$

(4) $\rightarrow$ (5) and simplifying:

$$(I + ml^2)\ddot{\theta} + ml\ddot{x}\cos\theta - mglsin\theta = 0 \tag{6}$$

Resulting Equations of Motion

So overall, our system of nonlinear equations of motion is:

$$(M+m)\ddot{x} + ml\ddot{\theta}\cos(\theta) - ml\dot{\theta}^2\sin(\theta) = F(t) \tag{7}$$

$$(I+ml^2)\ddot{\theta} + ml\ddot{x}\cos\theta - mglsin\theta = 0 \tag{8}$$

One may now rearrange this into a first order differential equation in state space form. It may be of interest to explore what the behaviour is around the equilibrium point ($\theta = 0$). To do this we may linearise the equations by using small angle approximations.

MATLAB Implementation of the system

```
%% Function: Nonlinear Dynamics
function dx = pendulum_dynamics(~, x, u, p)
    M = p.M; m = p.m; l = p.l; g = p.g; I = p.I;

    pos = x(1);
    vel = x(2);
    th = x(3);
    thd = x(4);

    denom = (M + m)*(I + m*l^2) - ((m*l*cos(th))^2);
    xdd = ( (I + m*(l^2))*(u + m*l*(thd^2)*sin(th)) - ((m^2)*(l^2)*g*cos(th)*sin(th)) ) / denom;
    thdd = ( (M+m) * (m*g*l*sin(th)) - (m * l * cos(th) * (u + m * l * (thd^2) * sin(th))) ) / denom;

    dx = [vel; xdd; thd; thdd];
end
```

Linearising with Small Angle Approximation

For small θ:

$$\cos(\theta) \approx 1, \sin(\theta) \approx \theta \tag{9}$$

So the resulting equations of motion become:

$$(M + m)\ddot{x} + ml\ddot{\theta} = F(t) \tag{10}$$

$$(I + ml^2)\ddot{\theta} + ml\ddot{x} - mgl\theta = 0 \tag{11}$$

General Systems

For a general nonlinear system:

$$\dot{x} = f(x, u) \tag{12}$$

And the equilibrium points being (x_e, u_e). We consider the Taylor expansion of $f(x, u)$ up to the first order derivative:

$$f(x, u) = f(x_e, u_e) + \left.\frac{\partial f}{\partial x}\right|_{x=x_e} (x - x_e) + \left.\frac{\partial f}{\partial u}\right|_{u=u_e} (u - u_e) \tag{13}$$

The partial derivatives $\left.\frac{\partial f}{\partial x}\right|_{x=x_e}$, and $\left.\frac{\partial f}{\partial u}\right|_{u=u_e}$ are matrices of partial derivatives of every component of f with every component of x, u respectively. The matrix is called the Jacobian.

Continued

We then define new variables of state, $\tilde{x} = x - x_e$ and $\tilde{u} = u - u_e$. Then we can write the equations of motion:

$$\dot{\tilde{x}} = A\tilde{x} + B\tilde{u} \tag{14}$$

where $A = \frac{\partial f}{\partial x}\Big|_{x=x_e}$, and $B = \frac{\partial f}{\partial u}\Big|_{u=u_e}$. Now with a linear approximation of the system, the tools developed for linear controller/observer design can be used (LQR, Pole placement, Linear MPC, etc).

```
%% Linearising the system

syms M m l g I pos vel th thd u
x = [pos; vel; th; thd];

% Copying the nonlinear eom
denom = (M + m)*(I + m*l^2) - (m*l*cos(th))^2;
xdd = ((I + m*l^2)*(u + m*l*thd^2*sin(th)) - (m^2*l^2*g*cos(th)*sin(th)))/denom;
thdd = ((M+m)*(m*g*l*sin(th)) - m*l*cos(th)*(u + m*l*thd^2*sin(th)))/denom;

f = [vel; xdd; thd; thdd];

% Computing the jacobian
A = jacobian(f, x);
B = jacobian(f, u);

% equilibrium
x_eq = [0; 0; 0; 0];
u_eq = 0;

% numeric parameter values
M_val = 1; m_val = 0.12; l_val = 0.5; g_val = 9.81; I_val = 0.01;

% symbolic variables as column vector
vars = [x; u; M; m; l; g; I];  % 9x1 column

% numeric values as column vector
vals = [x_eq; u_eq; M_val; m_val; l_val; g_val; I_val];  % 9x1 column

% now substitute
A_lin = double(subs(A, vars, vals));
B_lin = double(subs(B, vars, vals));

Co = ctrb(A_lin, B_lin);
Co
rank(Co)
```

Part 2: Introduction to PID Control

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On-Off Control

$$u = \begin{cases} u_{max} & \text{if } e > 0 \\ u_{min} & \text{if } e < 0 \end{cases} \tag{15}$$

- $e = r - y$
- Simplest control system
- Control variable oscillates - might be acceptable if small enough
- Control not defined when error is 0
- System overreacts to a small change in error

Proportional (P) Control

$$u = \begin{cases} u_{max} & \text{if } e \geq e_{max} \\ k_p e & \text{if } e_{min} < e < e_{max} \\ u_{min} & \text{if } e \leq e_{min} \end{cases} \tag{16}$$

- Controller input is proportional to the control error for small errors
- Might be enough
- Drawbacks: deviates from its reference value - prone to steady state error when there is a steady disturbance
- We want the error to go to 0 over time

Proportional-Integral Control (PI)

$$u = k_p e + k_I \int_0^t e(\tau) d\tau \tag{17}$$

- We want to track the error during the course of the system's motion
- Sum the error at each step - ie integration
- Drawbacks: steady-state may not be always present
- If we have magnitude limits - integrator windup can occur
- Careful design and analysis needed

Proportional-Integral-Derivative Control (PID)

$$u = k_p e + k_I \int_0^t e(\tau) d\tau + k_D \dot{e}(t) \tag{18}$$

- Derivative term increases the damping
- Provides information where the error is heading - linear extrapolation of the error
- Helps correct overshoot
- Prone to noise - which can result in controller being unstable, reducing robustness

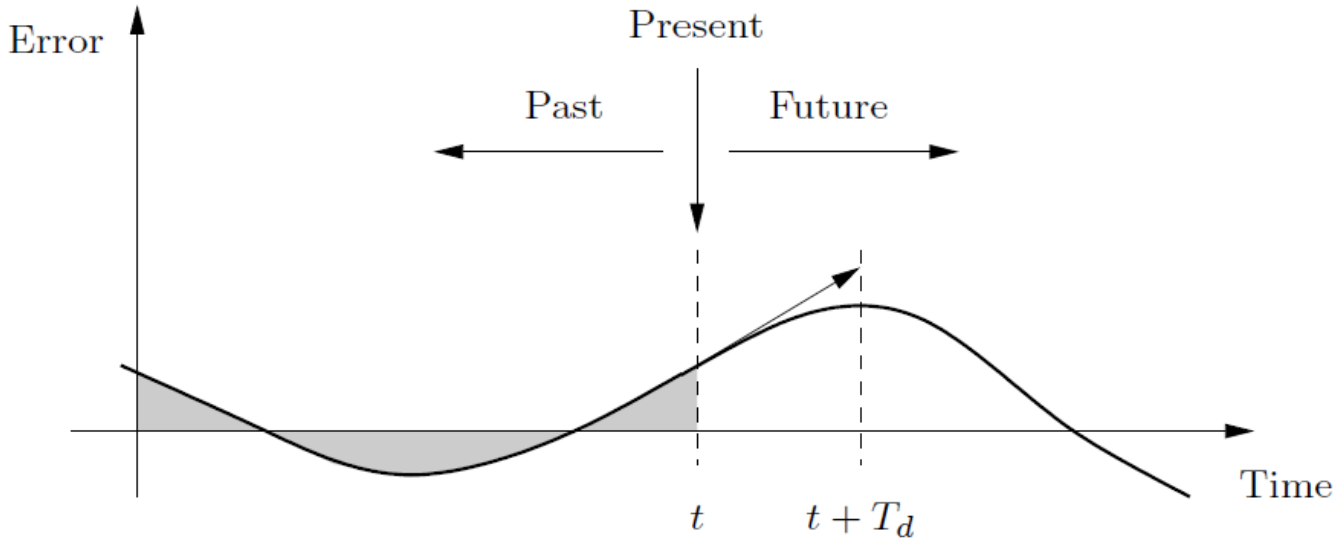


Figure 1.15: Action of a PID controller. At time t, the proportional term depends on the instantaneous value of the error. The integral portion of the feedback is based on the integral of the error up to time t (shaded portion). The derivative term provides an estimate of the growth or decay of the error over time by looking at the rate of change of the error. T_d represents the approximate amount of time in which the error is projected forward (see text).

TASK

- Modelling a 3DOF Thrust vectored rocket
- Obtain nonlinear equations of motion
- Linearise and obtain linear state space equation of motion
- Produce a Simulink diagram for it

Next 2 Weeks

- State Feedback
- Linear Quadratic Regulators
- Pole placement
- Controllability